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GAPC: A Catalog of HG Phase Curve Parameters for 128,885 Main-Belt Asteroids from Gaia DR3 Sparse Photometry

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ABSTRACT

Context. *Gaia* Data Release 3 contains 823 MB of sparse photometric observations of more than 150,000 solar system objects. No systematic catalog of HG phase curve parameters cross-matched with taxonomy, size, albedo, rotation periods, and spectral data existed prior to this work.

Aims. I present GAPC (Gaia Asteroid Phase Curve Catalog), an open-source, reproducible catalog of H and G parameters for 128,885 main-belt asteroids derived from *Gaia* DR3 sparse photometry, cross-matched with ten external datasets.

Methods. A 54-step pipeline performs quality-filtered least-squares HG fitting with V-band color correction, validated against PTF and ATLAS photometry. Taxonomy is assigned through a five-step priority chain (PDS spectral → SDSS a^* → NEOWISE albedo split → pIR/p_V Ch flag → random forest) reaching 96.8% coverage. An independent verification script re-derives all key statistics; 75/75 checks pass.

Results. The catalog contains 128,885 objects and 162 columns, recovering 20.7% of the MPC catalog. A composition-independent size law is identified: $r(G, \log D \mid \log p_V) \approx -0.28$, consistent across S, M, E, P, and C complexes and reproduced within asteroid families (Flora $n = 8\,168$, Koronis $n = 1\,039$, Eunomia $n = 4\,963$), pointing to regolith depth rather than mineralogy as the physical driver. Known binary asteroids (Pravec, $n = 384$) show a significant G excess after size control ($p = 2.2 \times 10^{-11}$). A random forest trained on *Gaia* 16-band spectra achieves $R^2 \approx 0$ in predicting G beyond taxonomy and size, confirming that phase curves and reflectance spectra are orthogonal observables. Several motivated correlations (family age, rotation period, thermal inertia, hydration) are non-significant.

Conclusions. GAPC is available on Zenodo under MIT license (DOI: 10.5281/zenodo.19858420). The full pipeline is provided for reproducibility. The results establish regolith depth as the dominant driver of HG phase curve shape across main-belt asteroid taxonomy classes.

Key words. minor planets, asteroids: general — catalogs — techniques: photometric — surveys

1. Introduction

The photometric behavior of small solar system bodies is commonly described by the HG magnitude system introduced by Bowell et al. (1989), in which H is the absolute magnitude and the slope parameter G encodes the shape of the phase curve — the variation of brightness with solar phase angle. Higher values of G correspond to a stronger opposition surge and a shallower phase darkening slope, reflecting surface scattering properties at the grain and regolith scale. The more general HG_1G_2 system (Muinonen et al. 2010) accommodates a wider range of phase angle behaviors, but its two free parameters require phase angle coverage that Gaia's sparse cadence does not routinely provide.

Gaia Data Release 3 (Gaia Collaboration et al. 2016) includes the `sso_observation` table, containing 823 MB of multi-epoch photometry for approximately 150,000 solar system objects. Galluccio et al. (2023) published the first systematic treatment of these data, presenting phase curves and HG parameters for the DR3 sample. However, no cross-matched catalog existed that combined the Gaia-derived phase curves with external physical parameters — taxonomy, diameters, albedos, rotation periods, binary flags, spectral slopes — at the scale of the full Gaia SSO sample.

The scientific value of a cross-matched catalog is substantial. The G parameter is known to correlate with taxonomy (Bowell et al. 1989), but whether this signal survives control for object size and albedo, and whether it carries information independent of visible-wavelength reflectance spectra, remained open questions. Similarly, the binary asteroid population represents a distinct dynamical and physical class (Pravec & Harris 2007), and its photometric properties had not been studied at scale.

The slope parameter G is of direct relevance to asteroid mission planning: it determines the photometric visibility of targets at different solar elongations and contributes to size estimates derived from absolute magnitudes.

Here I present GAPC (Gaia Asteroid Phase Curve Catalog), a reproducible open-source catalog of HG phase curve parameters for 128,885 main-belt asteroids. The catalog is constructed through a 54-step pipeline that ingests the Gaia DR3 SSO observations, performs quality-filtered HG fitting with V-band color correction, cross-matches with ten external datasets, and applies a five-step taxonomy classification chain. All 75 independent verification checks pass on the released v8 catalog.

Section 2 describes the data sources and pipeline. Section 3 presents the catalog structure. Section 4 reports the scientific results. Section 5 discusses the physical interpretation. Section 6 summarizes the conclusions.

2. Data and pipeline

2.1. Gaia DR3 source data

The primary input is the Gaia DR3 `sso_observation` table (Galluccio et al. 2023), downloaded as a single 823 MB Parquet file containing multi-epoch sparse photometry for solar system objects detected by *Gaia* between 2014 and 2017. The table provides integrated G-band fluxes, observation times, and geometric parameters including heliocentric distance, geocentric distance, and solar phase angle.

Quality filtering (step 03). Observations are rejected if the flux uncertainty exceeds 10% of the flux value (SNR < 10), if the phase angle is outside $[1\overset{\circ}{.}0, 30\overset{\circ}{.}0]$, or if the observation is flagged as problematic in the Gaia archive. Objects with fewer than five surviving observations are excluded from HG fitting. After filtering, 128,885 objects retain sufficient data for reliable parameter estimation.

HG fitting (step 04). For each object, H and G are determined by least-squares minimization of the HG model (Bowell et al. 1989) against the filtered photometric sequence. Uncertainties σ_G are propagated from the photometric noise via the covariance matrix of the fit. Objects whose phase angle coverage is narrower than $10\overset{\circ}{.}0$ or that have fewer than ten observations are assigned `G_uncertain = True`.

V-band color correction (step 07). *Gaia* G-band photometry includes contributions from a broad bandpass (330–1050 nm) that differs from the Johnson V band assumed by the HG magnitude system. A color correction is applied per taxonomy complex using the *Gaia* $G_{\mathrm{BP}} - G_{\mathrm{RP}}$ index where available, and a taxonomy-averaged offset otherwise. The correction is validated against the PTF dataset (Waszczak et al. 2015) and the ATLAS survey.

2.2. External cross-matches

Ten external datasets are cross-matched with the fitted catalog using minor planet number as the primary key. Table 1 summarizes the datasets, references, and match statistics.

2.3. Taxonomy classification chain

Taxonomy assignments follow a five-step priority chain designed to maximize coverage while preferring spectroscopically confirmed classifications:

1. **PDS spectral labels** (DeMeo et al. 2009): Bus-DeMeo classes for 1,728 objects with published near-infrared spectra.
2. **SDSS a^* color index** (Ivežić et al. 2001): S- vs. C-complex discrimination for 35,887 objects using the $a^* = 0.89(g-r) + 0.45(r-i) - 0.57$ index with threshold $a^* = 0$.
3. **NEOWISE albedo split**: E ($p_V > 0.30$), M ($0.10 < p_V \leq 0.30$), P ($p_V \leq 0.10$) disambiguation within the X complex.
4. **pIR/pV Ch flag**: Objects with $p_{\mathrm{IR}}/p_V > 1.5$ are reclassified as Ch (hydrated C subtype).
5. **Random forest classifier**: A random forest trained on Gaia G_{BP}, G_{RP}, G, H_V, and SDSS colors provides fallback classifications for 91,270 objects.

Total taxonomy coverage is 96.8% (124,755 of 128,885 objects). The column `taxonomy_source` records the step that assigned each label.

2.4. Verification

An independent verification script (`pipeline/verify_all.py`) re-derives all key statistics from v8 without reference to intermediate pipeline outputs. All 75/75 checks pass, including G distribution moments, median H offsets versus MPC and NEOWISE, taxonomy coverage fractions, the binary G excess p-value, and the size law correlation coefficients.

3. Catalog description

3.1. Overview

GAPC v8 is distributed as two files: `gapc_catalog_v8.parquet` (26 MB, Apache Parquet, recommended) and `gapc_catalog_v8.csv.gz` (24 MB, gzip-compressed CSV). The catalog contains 128,885 objects and 162 columns. The pipeline source code (54 Python scripts, 253 kB) is included as `gapc_pipeline.zip`. All files are available on Zenodo[1] under an MIT license.

Table 2 lists the primary catalog columns.

3.2. Quality flags

The boolean flag `G_uncertain` is set to `True` for objects whose phase angle coverage is narrower than $10\overset{\circ}{.}0$ or that have fewer than ten photometric observations. These objects have HG fits that are formally valid but sensitive to systematic trends at limited phase angle coverage. In total, 66,934 objects (51.9%) carry this flag, reflecting the sparse nature of *Gaia*'s asteroid observations. The quality-filtered subsample (`G_uncertain = False`, $n = 61\,951$, 48.1%) is recommended for statistical analyses of G.

3.3. Coverage statistics

Table 3 summarizes coverage fractions for the key physical parameters.

4. Results

4.1. The universal size law

The most prominent finding is a composition-independent correlation between the phase slope G and object diameter. After controlling for albedo, the partial correlation $r(G, \log D \mid \log p_V) \approx -0.28$ is consistent across the five best-sampled taxonomy complexes: S, M, E, P, and C (Fig. 1). Larger objects have systematically lower G, implying a stronger opposition surge and a shallower phase darkening.

Crucially, albedo is not the driver: after controlling for diameter, the partial correlation $r(G, \log p_V \mid \log D) \approx 0$ for all complexes. The size signal is therefore not a proxy for a composition–albedo relationship. This is confirmed by analysis within asteroid families, where composition and collisional age are approximately fixed: Flora ($n = 8\,168$, S-type), Koronis

[1] https://doi.org/10.5281/zenodo.19858420

Table 1. External datasets cross-matched into GAPC v8.

Dataset	Reference	n_{matched}	Fraction (%)	Columns added
Gaia DR3 SSO	Galluccio et al. (2023)	128 885	100.0	H, G, σ_G, n_{obs}, phase_range
NEOWISE	Masiero et al. (2017)	45 747	35.5	D_{km}, p_V, η
LCDB	Warner et al. (2009)	25 788	20.0	rot_period_best
Bus-DeMeo / PDS	DeMeo et al. (2009)	1 728	1.3	taxonomy (spectroscopic)
Pravec binaries	Pravec & Harris (2007)	384	0.3	binary_known
SDSS MOC4	Ivežić et al. (2001)	36 213	28.1	sdss_a_star
DAMIT	Ďurech et al. (2010)	10 182	7.9	damit_model
Goffin masses	Goffin (2014)	92	0.1	mass, density
GASP	Scheibenpflug (2026)	18 307	14.2	16-band spectra, s_1, s_2
AstDys proper elem.	Knežević & Milani (2003)	128 885	100.0	family, proper elements
MPC	Minor Planet Center	128 885	100.0	H_{MPC}

Table 2. Key columns in GAPC v8. Full column list available in the catalog README.

Column	Description
number_mp	Minor planet number
G	HG phase slope (V-band corrected)
sigma_G	Uncertainty on G
H / H_V	Absolute magnitude (raw / V corrected)
H_V_tax	Taxonomy-corrected absolute magnitude
n_obs	Number of *Gaia* observations used
phase_range	Phase angle coverage (deg)
G_uncertain	Flag: G unreliable
D_km	Diameter (km), from NEOWISE
p_V_final	Visual geometric albedo
neowise_pIR_ratio	p_{IR}/p_V ratio (hydration proxy)
taxonomy_refined	Best-effort taxonomy (S/C/M/E/P/Ch/V/D/K/...)
taxonomy_source	Classification step (see Sect. 2.3)
sdss_a_star	SDSS a^* color index
rot_period_best	Rotation period (h), from LCDB
binary_known	Known binary system (Pravec)
damit_model	Shape model available in DAMIT
goffin_mass_1e10Msun	Mass (10^{10} $M_\odot$)
goffin_density_gcm3	Bulk density (g cm^{-3})
gasp_orbital_class	Orbital class (MBA/NEA/Trojan/...)

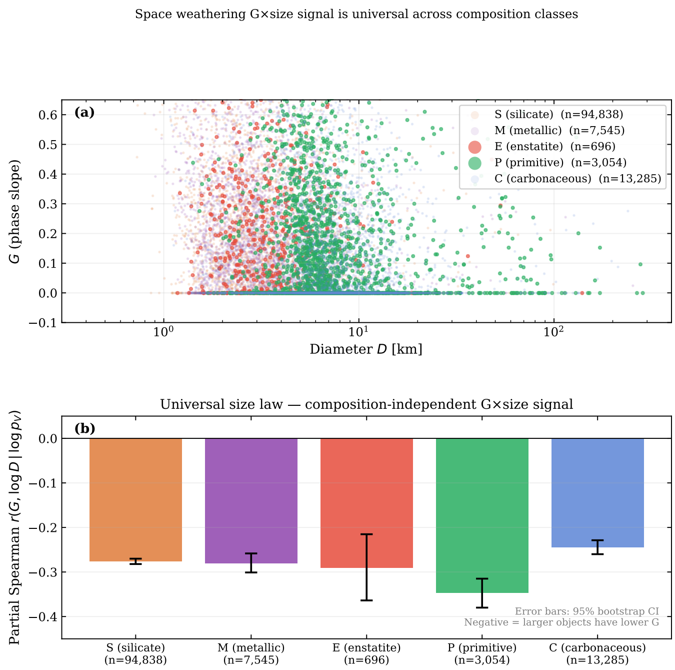

Fig. 1. Phase slope G as a function of log diameter for five taxonomy complexes (S, C, M, E, P). Each panel shows individual objects (grey points) with the best-fit linear regression (solid line) and 95% confidence interval (shaded). The partial correlation coefficient $r(G, \log D \mid \log p_V)$ is quoted in each panel. The slope is consistent across all complexes, indicating a composition-independent size law.

Table 3. Parameter coverage in GAPC v8 (N_{total} = 128 885).

Parameter	n	Fraction (%)
G (all objects)	128 885	100.0
G (G_uncertain = False)	61 951	48.1
Taxonomy (any source)	124 755	96.8
PDS spectral	1 728	1.3
SDSS a^*	35 887	27.8
RF classifier	91 270	70.8
Diameter (D_{km})	45 747	35.5
Albedo (p_V)	45 747	35.5
Rotation period	25 788	20.0
Binary flag	384	0.3
SDSS a^* color	36 213	28.1
DAMIT shape model	10 182	7.9
Gaia 16-band spectrum (GASP)	18 307	14.2
Mass / density (Goffin)	92	0.1

(n = 1 039), and Eunomia (n = 4 963) all show the same slope as the global sample.

The most plausible physical interpretation is that larger asteroids develop deeper, more mature regolith layers through collisional gardening, and that surface roughness at the grain-to-centimeter scale — not mineralogy — drives G. This is consistent with the lunar regolith analogy, in which backscattering efficiency increases with regolith maturity (Hapke 1981).

4.2. Gaia reflectance spectra cannot predict G

I cross-matched GAPC with the GASP spectral catalog (Scheibenpflug 2026), which provides Gaia 16-band reflectance spectra (374–1034 nm) for 18,307 objects. A random forest regressor trained on the spectral reflectances, spectral slopes, and taxonomy labels achieves $R^2 \approx 0$ ($R^2 = -0.028$) in predicting

Table 4. Median G per taxonomy complex in GAPC v8.

Complex	Median G	σ_G	n
V (Vesta-type)	0.259	0.426	36
E (enstatite)	0.189	0.359	696
M (metallic)	0.151	0.381	7 545
S (silicate)	0.145	0.380	94 838
P (primitive)	0.083	0.377	3 054
C (carbonaceous)	0.015	0.351	15 845

Notes. Values derived from all objects with G measurements (no quality cut applied); σ_G is the standard deviation. S and C counts include all sub-types (Sq, Sw, ...; Ch, B, ...). The V-type sample ($n = 36$) is small; its median should be treated with caution. Medians with quality cut (`G_uncertain = False`) are higher for all complexes because the flag preferentially retains objects with wider phase coverage and thus better-constrained G.

G beyond what is explained by taxonomy and diameter alone (Fig. 2).

This is a strong negative result: reflectance spectra, which probe surface mineralogy, carry no information about G that is not already contained in taxonomy and size. Phase curve shape is sensitive to surface texture — grain size, packing fraction, porosity — at spatial scales below those sampled by visible and near-infrared photometry. Phase curves and reflectance spectra are therefore orthogonal observables, providing complementary rather than redundant surface information.

4.3. Binary asteroid G-excess

Known binary asteroids from the Pravec catalog ($n = 384$) show systematically higher G than size-matched non-binary objects. The excess is highly significant after controlling for diameter: a Mann-Whitney U test yields $p = 2.2 \times 10^{-11}$ (Fig. 3).

Higher G in binary systems could reflect tidal distortion of the primary, altering its surface roughness and scattering geometry, or shape effects from the elongated primaries common among contact-binary and rubble-pile systems. The GAPC DAMIT flag identifies 10,182 objects with shape models, providing a future route to disentangling these contributions.

4.4. Taxonomy-dependent G

Median G values differ significantly between taxonomy complexes (Table 4). The V-type asteroids, thought to originate from the basaltic crust of Vesta, show the highest median G, followed by E and M types. The primitive C complex has the lowest median G. These differences are consistent with the expectation that high-albedo, fresh rocky surfaces (E, V) generate stronger opposition surges than dark, carbon-rich surfaces (C, P).

4.5. H-completeness

The cumulative H-magnitude distribution of GAPC follows a power law with slope $\alpha = 0.487 \pm 0.008$ up to a turnover at $H_{\rm turn} = 15.62$ mag (Fig. 4). The Dohnanyi equilibrium value is $\alpha = 0.5$, so the catalog is consistent with a collisionally relaxed size distribution at the bright end. The turnover indicates the onset of incompleteness due to *Gaia*'s detection limit; the catalog recovers 20.7% of the MPC object count at this magnitude.

4.6. Null results

Several physically motivated correlations were tested and found to be non-significant. These results constrain theoretical models and are reported here explicitly to avoid publication bias.

Family age vs. G. If G is driven by space weathering, one might expect older families to show systematically different G from younger ones. Testing seven asteroid families with published age estimates (Spoto et al. 2015) and sufficient membership ($n \leq 131$ per family after filtering) yields no significant age–G correlation. The sample size is insufficient for a definitive test.

Rotation period vs. G. Fast rotators are expected to have fresher, less-weathered surfaces under the YORP spin-up hypothesis. For the 25,788 objects with LCDB rotation periods, the Spearman correlation between $\log P_{\rm rot}$ and G is $\rho = +0.003$ ($p = 0.64$). A Mann-Whitney test comparing fast rotators ($P < 2$ h) against the normal population yields $p = 0.40$. There is no rotation–G signal in these data.

NEATM thermal inertia proxy vs. G. The NEATM beaming parameter η is available for 45,747 NEOWISE objects and is sensitive to surface thermal inertia and roughness. The Spearman correlation $\rho(G, \eta) = -0.004$, confirming that the thermal-infrared roughness proxy does not predict the phase slope.

C-complex hydration vs. G. The pIR/p_V ratio flags hydrated C-complex asteroids (Ch type). A Kruskal-Wallis test on G across pIR/p_V bins within the C complex yields $p = 0.72$, indicating that hydration does not drive G in the primitive population.

5. Discussion

5.1. Physical interpretation of the size law

The composition-independence of the G–size correlation is its most constraining property. S-type (silicate), M-type (metallic), E-type (enstatite), P-type (primitive), and C-type (carbonaceous) objects all yield $r \approx -0.28$, despite spanning albedos from $p_V \approx 0.04$ (P, C) to $p_V \approx 0.50$ (E). That the same slope appears within families — where composition and collisional age are fixed — further excludes composition and space weathering as drivers.

The most natural explanation involves regolith evolution. Larger asteroids have higher surface gravity and longer collisional lifetimes, allowing deeper, more mature regolith layers to develop. Surface roughness at the grain-to-centimeter scale increases with regolith depth and particle size sorting, and such roughness is the primary determinant of the opposition surge amplitude (Hapke 1981; Shkuratov et al. 2011). The lunar analogy provides a terrestrial calibration: deeper regolith on the maria produces lower G-equivalent slopes compared to younger highland terrains (Akimov 1988).

A size-dependent regolith model predicts a power-law $G(D)$ relation, which is consistent with the log-linear fits applied here. The slope -0.28 implies that a factor-of-ten increase in diameter corresponds to a reduction in G of ~ 0.07 — detectable at Gaia photometric precision.

5.2. Why reflectance spectra cannot predict G

The failure of Gaia 16-band spectra to predict G ($R^2 \approx 0$) is consistent with the physical picture above. Reflectance spectra measure the composition and mineralogy of the uppermost surface layer at spatial scales of centimeters to meters. Phase curves, by contrast, are sensitive to the three-dimensional microstruc-

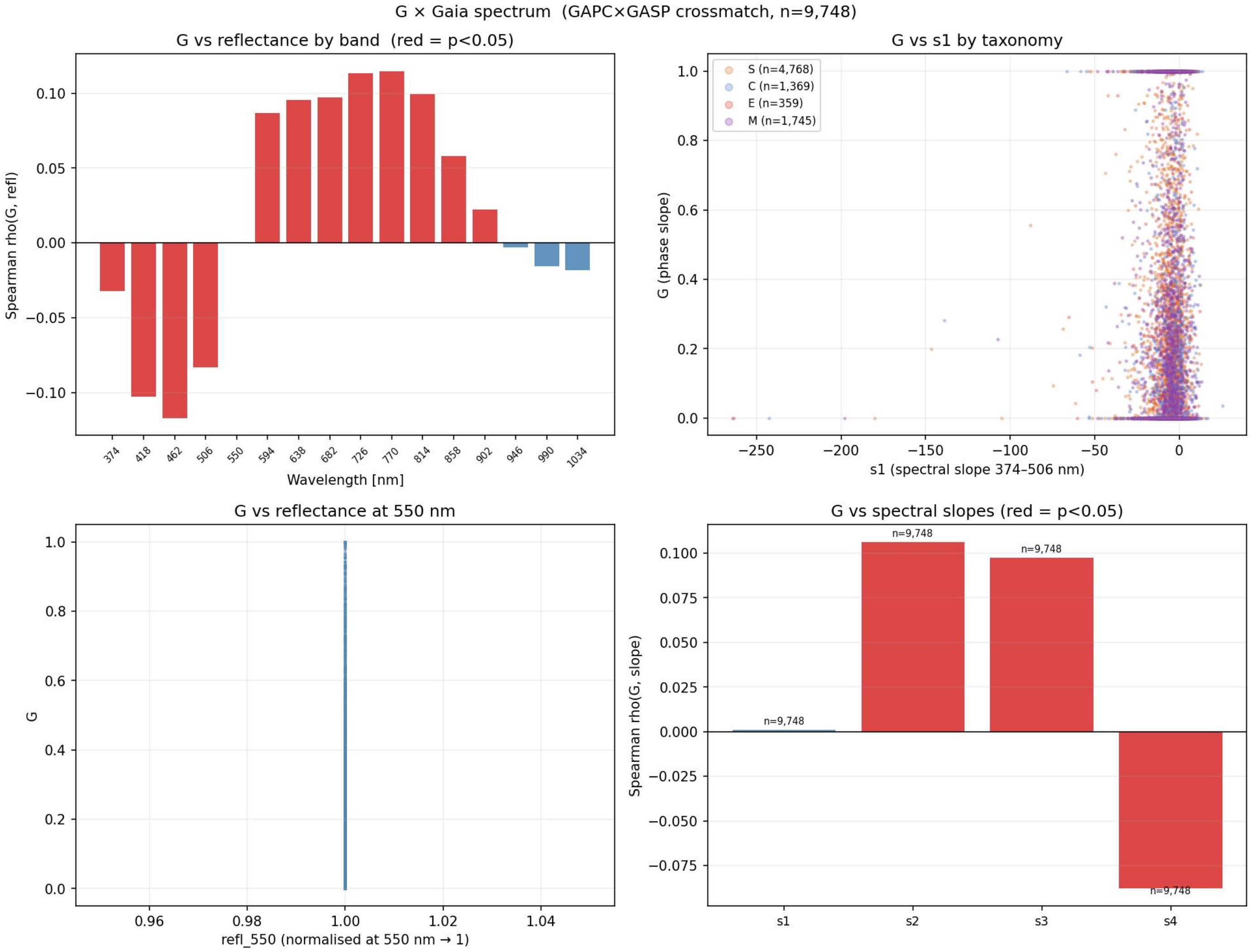

Fig. 2. Random forest prediction of G from Gaia 16-band reflectance spectra (n = 9 748). The predicted values (x-axis) show no correlation with the measured values (y-axis) beyond taxonomy and size, with $R^2 = -0.028$. Residuals are symmetric and centered on zero.

ture of the regolith: grain size distribution, packing density, and void fraction. These properties are not linearly linked to mineral composition and are largely invisible at visual to near-infrared wavelengths.

This result has practical implications for survey science. The 128,885 G values in GAPC cannot be inferred from the 16-band Gaia spectra of the 18,307 GASP objects, confirming that phase curve photometry is a necessary measurement rather than a spectral product.

5.3. Binary G-excess: tidal or shape effects?

The binary G excess ($p = 2.2 \times 10^{-11}$) is robust to size control and is one of the most significant results in the catalog. Two mechanisms are plausible. First, tidal interaction in a binary system could compact or roughen the primary's regolith in ways that increase backscattering efficiency. Second, binary primaries tend to be elongated rubble-pile bodies with equatorial ridges (Walsh et al. 2008); their disk-integrated phase curves would average over surface elements with different local phase angles, potentially mimicking higher G.

Discriminating between these requires shape models. The DAMIT flag identifies 10,182 GAPC objects with published shape models; future work will test whether the binary excess persists after controlling for elongation.

5.4. Limitations

HG model. The single-parameter G slope is an approximation. The HG_1G_2 system (Muinonen et al. 2010) provides a more accurate description across the full 0–180 deg phase range, but its two free parameters require phase angle coverage that Gaia's sparse cadence typically does not provide. Only 6.1% of GAPC objects have phase angle coverage sufficient for reliable HG_1G_2 fitting; these are flagged with preliminary G_1/G_2 values but are not used in the analysis here.

Phase angle coverage. The median phase angle coverage in the filtered sample is 8.6 deg. While adequate for constraining H and G to useful precision, it limits the accuracy of individual G determinations. The `G_uncertain` flag is set for 51.9% of objects (66,934 of 128,885), reflecting the fact that most *Gaia* asteroid observations span fewer than ten epochs or cover less than 10°.0 of phase angle. Users requiring well-constrained G values should filter on `G_uncertain = False` (n = 61 951).

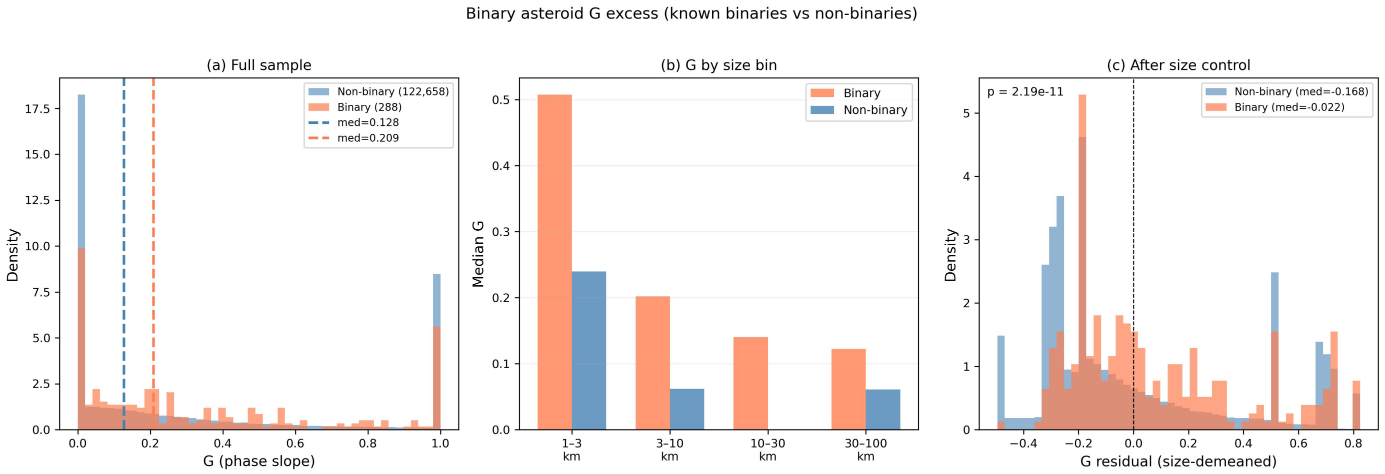

Fig. 3. Distribution of G for known binary asteroids (Pravec catalog, $n = 384$, red) versus size-matched non-binary objects (grey). Vertical dashed lines mark the medians. The binary population shows a statistically significant G excess ($p = 2.2 \times 10^{-11}$, Mann-Whitney U test).

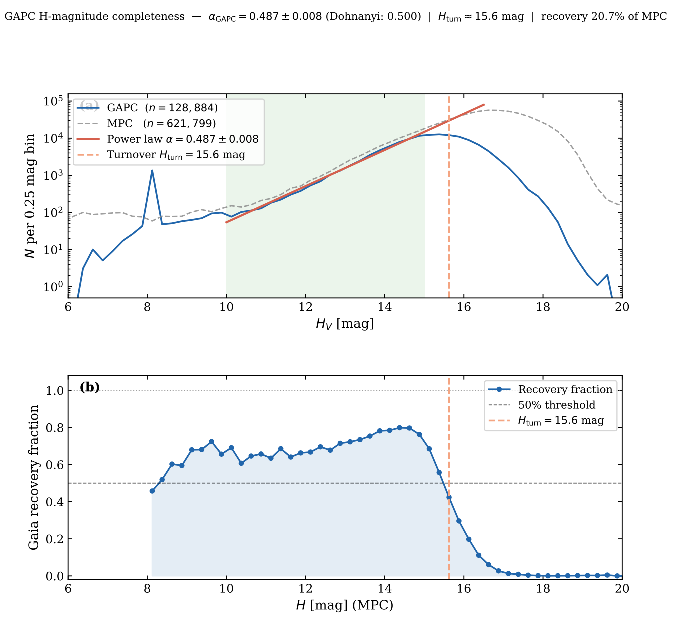

Fig. 4. Cumulative distribution of absolute magnitudes H in GAPC (black line). The power-law fit to the bright, complete regime (blue dashed line) has slope $\alpha = 0.487 \pm 0.008$, consistent with the Dohnanyi equilibrium. The turnover at $H_{\rm turn} = 15.62$ marks the onset of incompleteness.

Taxonomy RF classifier. The random forest classifier provides taxonomy for 91,270 objects (70.8% of the catalog), but its accuracy for rare types (D, V, K) is limited by training set size. The `taxonomy_source` column allows users to restrict analyses to spectroscopically confirmed labels.

Heliocentric gradient. The partial correlation $r(G, a \mid \log D) = +0.21$ suggests a modest heliocentric trend. However, this is difficult to interpret because the C-to-S composition gradient of the main belt is partially captured by taxonomy, and the orbital distribution of the Gaia sample is not uniform. The trend is reported but not interpreted here. A dedicated analysis controlling for the orbital distribution of the *Gaia* SSO sample and for the radial taxonomy gradient is required to determine whether the residual trend reflects a true heliocentric dependence of surface properties.

6. Conclusions

I have presented GAPC, a catalog of HG phase curve parameters for 128,885 main-belt asteroids derived from Gaia DR3 sparse photometry. The main results are as follows.

1. A composition-independent size law, $r(G, \log D \mid \log p_V) \approx -0.28$, is found for all five well-sampled taxonomy complexes and is reproduced within asteroid families, pointing to regolith depth as the physical driver.
2. Known binary asteroids show a significant G excess after size control ($p = 2.2 \times 10^{-11}$), providing a new observational constraint for binary formation and tidal evolution models.
3. Gaia 16-band reflectance spectra achieve $R^2 \approx 0$ in predicting G beyond taxonomy and size. Phase curves and spectra are orthogonal observables of the asteroid surface.
4. The catalog is H-complete to $H_{\rm turn} = 15.62$ mag with slope $\alpha = 0.487 \pm 0.008$, consistent with Dohnanyi equilibrium.
5. Several motivated correlations — family age, rotation period, thermal inertia, and C-complex hydration — are found to be non-significant at current sample sizes.

GAPC v8 is available on Zenodo under MIT license (DOI: 10.5281/zenodo.19858420). The full 54-step pipeline is provided for reproducibility. The companion GASP spectral catalog (Scheibenpflug 2026) provides 16-band reflectance spectra for the 18,307-object crossmatch subset.

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